Genetics of Toxin Production and Prediction

CyanoDTec

Phytoxigene™ CyanoDTec a quantitative Real Time PCR assay for the detection and quantitation for the presence of Cyanobacteria and their toxin producing genes from aquatic environmental samples.

Catalogue Numbers

205-0050 **Phytoxigene™ CyanoDTec Total Cyanobacteria**; (16S rRNA) and Internal Amplification Control (IAC) target

205-0051 **Phytoxigene™ CyanoDTec Toxin Gene**; microcystin/nodularin, cylindrospermospin and saxitoxin

CyanoNAS **Phytoxigene™ CyanoNAS;** Nucleic Acid Standards for the CyanoDTec kit.

INTRODUCTION;

Phytoxigene™ CyanoDTec is a molecular test based on quantitative Real Time PCR. The assay detects and quantifies the presence of cyanobacteria (blue green algae) and their toxin producing genes from aquatic environmental samples. Not all cyanobacteria species produce toxins, therefore the presence of a cyanobacteria does not immediately defer a risk of toxins being present. The Phytoxigene test quantitates both the amount of overall Cyanobacteria present in a sample along with the number of genes that are responsible for the production of the certain toxins. Toxins associated with blue green algae can be hepatatoxins or neurotoxins. The hepatatoxins include microcystin, nodularin and cylindrospermopsin while saxitoxin is one of the primary neurotoxins produced by cyanobacteria.

KIT CONTENTS;

The **Phytoxigene™ CyanoDTec** assay is made up of two separate tests, each with its own set of reagents. They are packaged in two separate aluminium foil pouches. The Individual tests can be purchased separately depending on the application and algorithm chosen by the laboratory, but it is advised that at a minimum the Total Cyanobacteria assay be run on each sample to insure that the operator identifies any possible inhibition with the assay by utilising the Internal Amplification Control (IAC) contained within the Total Cyanobacteria test.

Product #	Description and Target	Contents	# Tests
205-0050	**Phytoxigene™ CyanoDTec Total Cyanobacteria**; (16S rRNA) and Internal Amplification Control (IAC) target	6 tubes (green cap) each containing all necessary reagents (enzymes, probe, primers and dNTP) for 4 tests	24
205-0051	**Phytoxigene™ CyanoDTec Toxin Gene**; microcystin/nodularin, cylindrospermospin and saxitoxin	6 tubes (blue cap) each containing all necessary reagents (enzymes, probe, primers and dNTP) for 4 tests	24

REAGENTS AND CONSUMABLES REQUIRED BUT NOT SUPPLIED WITH KIT;

Phytoxigene™ CyanoNAS are Nucleic Acid Standards produced for the CyanoDTec assay by the National Measurement Institute, Australia. The standards are certified concentrations of the target genes used in the CyanoDTec test. There are 5 standards in the CyanoNAS kit 100, 1,000, 10,000, 100,000 and 1,000,000 copies per reaction for each target; 16S rRNA gene of Cyanobacteria, microcystin and nodularin (mycE/ndaF), cylindrospermopsin (cyrA) and saxitoxin (sxtA). Product Catalogue is: # CyanoNAS. It can be purchased from Phytoxigene.

PCR grade water for the rehydration of the lyophilised beads in each tube. It can be purchased through a number of suppliers.
BioGx Bead Lysis tubes, contains 500µl of lysis buffer along with 300mgs of 0.1um glass beads; Product catalogue #800-1000. It can be purchased from Phytoxigene.
25 or 47mm diameter filters with maximum pore size of 3 micron, and can be purchased through a number of suppliers
PCR Reaction plates/tubes/strips, Dependent on PCR system utilised; plates, tubes or microtubes for the PCR Reaction

EQUIPMENT REQUIRED FOR PERFORMING THE ASSAY:

- Real Time PCR System capable of reading the relevant dye sets used in the assay. The **Phytoxigene™ CyanoDTec** assay has been primarily validated for use on the Cepheid Smartcycler™ instrument but other systems have been tested and shown to be adequate for the purpose of running the **Phytoxigene™ CyanoDTec** assay.
- Filtration system (syringe with reusable filter holder or flow through cup device with vacuum manifold) suitable for concentrating sample
- Bead Beater or Vortex Shaker with horizontal microtube adapter
- Microtube centrifuge
- 5µl pipette or equivalent
- 20-200µl variable pipette
- 200µl-1ml variable pipette

SAMPLE COLLECTION, SHIPMENT AND PRESERVATION;

1. Volume collected: Up to 250ml, minimum 10ml.
2. Sample Container: glass or PETG (Nalgene)
3. Collection & Shipment:
    a. Rinse container with sample then fill container, leaving enough headspace to allow for mixing. For grab samples, if no surface scum is present, collect 6”- 8” below the surface. For composite samples, make certain collection near the bottom does not disturb sediment which may cause assay inhibition.
    b. Samples must be kept sealed after collection and its recommended that it be kept at a temperature not exceeding 10°C during shipment to the laboratory.
4. Preservation & Processing:
    a. Sample temperature to be stored at or below 6°C at the laboratory. Samples can come to room temperature during processing (under two hours).
    b. Samples must be filtered for extraction within 48 hours of collection. Before filtration, thoroughly mix the sample. Always log volume filtered for quantification purposes. After filtration the samples should be extracted and analysed on the same day. If this is not possible, store the filters and/or extracts at -20°C.

SAMPLE PREPARATION AND LIMITS OF DETECTION;

The assay uses extracted DNA obtained from the filtration of a primary water sample, followed by the mechanical and/or chemical lysing of the cells. The quality of the sample preparation has significant impacts on the quality of the overall result. Each of the recommended methods has its merits and limitations and is dependent on the application sort after by the operator. For ongoing monitoring of water system’s the standardisation of this procedure is critical.

The starting template or sample size for the assay is five (5)µl. The limit of detection for the assay, with a 95% degree of confidence is 45 copies. Samples that are positive but below this level will likely be reported as positive by the instrument, they have no value assigned to them. They should be reported as “Below Detectable Limits “(BDL). With that in mind the starting volume of the primary sample will have considerable impact on the overall limit of detection of the assay when reporting copy number per ml. The follow chart outlines the effect of this limit of detection and its dependency to input volume

Primary Sample Volume	Assay Limit of Detection in assay	Minimum copy number detected in 1ml with 95% confidence level
500µl direct sample with 500µl lyse buffer/beads	45	9,000 copies per ml
5ml concentrated sample with 500µl lyse buffer/beads	45	900 copies per ml
10ml concentrated sample with 500µl lyse buffer/beads	45	450 copies per ml

Dependent on the needs of the operator, different starting volumes can be utilised. For obvious and known cyanobacteria scenarios where the levels are reasonably high and the objective is to determine whether toxin genes are present at levels higher than the limit of detection, a small sample can be directly lysed and processed. In monitoring programs where sensitivity is required it is recommended to start with a minimum of 10ml primary sample and concentrate by filtration followed by mechanical and chemical lysing using a bead beating methodology. A traditional spin column DNA extraction kit suitable for bacteria or water can be utilised after the lysing step to obtain higher yields should this be required.

Caution. Samples should be processed as soon as possible. If samples are being batched and not run on the assay within a couple of days, the samples should minimally be filtered or centrifuged, depending on the chosen method of preparation and then stored frozen until ready to process to the next step. Degradation of the sample, especially the viability of the cyanobacteria is affected when stored in sealed containers for long periods. Toxin gene concentrations can decline markedly when the sample is stored at room temperature for extended periods, ideally samples need to be filtered and processed within 2 days from collection. It is not recommended to store samples at minus 20°C before processing, the cells may lyse and DNA lost during subsequent filtration.

Suggested options for producing the starting template for the PCR reaction.

- For obvious and high-level blooms or applications where sensitivity is not critical;

  Mix sample by shaking container back and forth for 10 seconds, immediately pipette 500µl into a BioGx Bead Lysis Tube or equivalent and beat by using a dedicated mini bead beater (BioSpec Corp. /Omni Bead Ruptor/ MP Bio FastPrep-24™or equivalent) for 2 min at its highest speed setting. Conversely place the BioGx Bead Lysis tube onto a horizontal microtube holder (Scientific Industries SI-H524 or equivalent) attached to a heavy duty vortexer (Scientific Industries Genie 2 or equivalent) for a minimum of 15 minutes at its highest setting. After bead beating, centrifuge tubes using the microtube centrifuge for 1 min to pellet the beads and debris. Transfer as much of the supernatant to another microcentrifuge tube without disturbing the pellet, typically 400-450 µl. Pulse-vortex the transferred supernatant prior to loading into the PCR reaction tube or well.

- Regular monitoring and standard sample preparation

  A. Filter 5 to 100 ml of sample (25ml volume is most typical) through a vacuum filter holder assembly using a 25 or 47mm diameter, 0.8 to 3-µm pore size polycarbonate or equivalent filter (e.g. Pall Versapor filter cat# 66393, 25mm 1.2um; cat# 66400, 25ml 0.8um). Vacuum until there is no visible moisture. Alternatively utilise a reusable filter holder assembly (e.g. Pall 25mm In-line Delrin Plastic filter holder, cat#1109) and a disposable syringe. Syringe sample through the filter housing after inserting the appropriate filter into the holder.

     In both methods remove the filter from the manifold with a clean or disposable forceps, gently fold or roll the filter and place into the bead beating tube, filtrate side up facing inwards into the tube. Always log volume filtered for quantification purposes. Continue by following the same procedure for beating the cells using the BioGx Bead Lysis tubes as outlined in Method 1.

B. The Ohio EPA has made regular use of the assay since 2016 and their protocols for sample collection & preparation are covered in their SOP 705.0 Version 1.0 Sections 4-9. https://epa.ohio.gov/Portals/28/documents/habs/705.0-qPCR.pdf

NOTE: If using the reusable filter holders the holders will need to be cleaned between samples, the housing and seals need to be rinsed and soaked. Minimally a series of 2 rinses using water. The device is also suitable for autoclaving.

- High Yield and Sensitivity

  After completing method 2, utilise a specific water/soil bacterial DNA extraction kit eg MoBio **PowerWater® DNA Isolation Kit** , **RapidWater® DNA Isolation Kit** or equivalent, follow procedures as laid out in the package insert;

  NOTE: at any time a sample extracted using method 1 or 2 is shown to be inhibited during the amplification process, wherein the Internal Amplification Control (IAC) contained in the Total Cyanobacteria test returns a CT value 1.5 cycles higher than the CT of a blank or Non Template Control sample, the sample either needs to be diluted (minimally 1:10) or run through a purification kit (see Method 3) and run again.

PROCEDURE;

The **Phytoxigene™ CyanoDTec test** is performed either in parallel or sequentially using the 2 individual pouch components that make up the test. Running the Total Cyanobacteria 16s rRNA/IAC assay first allows for the interpretation of the presence, absence and quantity of cyanobacteria and whether the sample contains any inhibitors that would invalidate the assay. By using this approach it may not be necessary to run the Toxin Gene assay, especially in the early part of the monitoring season when background cyanobacteria levels are low. Performing the test in parallel saves time, especially when it's clear there is cyanobacteria present.

In each pouch there are 6 tubes of lyophilised reagents capable of 4 tests, therefore each pouch contains enough material for 24 tests. Each tube contains all the reagents necessary for the assay.

The Phytoxigene™ CyanoNAS standards each containing 100µl of each level, or 20 reactions of 5 µl each. The standards are not supplied with the kit and need to be ordered separately.

Standard (cap colour)	Copies/µl	Copies per reaction (5µl)
NA012 (green)	20 copies/µl	100 copies/reaction
NA013 (yellow)	200 copies/µl	1,000 copies/reaction
NA014 (blue)	2,000 copies/µl	10,000 copies/reaction
NA015 (orange)	20,000 copies/µl	100,000 copies/reaction
NA026 (pink)	200,000 copies/µl	1,000,000 copies/reaction

The CyanoNAS standards are shipped on cold packs, upon receipt the standards should be kept at -20°C. **After first use, the standards are recommended to be stored at 4-8°C, to avoid frequent freeze/thaw cycles**. **Standards can be kept at 4-8°C for up to 6 months**.

Before first use, after removal from freezer, bring to room temp (minimally 15 minutes). To mix, either flick each tube several times or pulse vortex each. Then pulse spin before pipetting.

Each tube in each of the respective test pouch (Total Cyanobacteria/Toxin Gene) is reconstituted with 80µl of PCR grade water. Rehydrate enough tubes to cover the requirements for the number of tests that being performed at any one time. After rehydration vortex each tube followed by a brief spin. If performing more than 4 tests, combine the volume from each tube of mastermix into a single vial. To ensure that there is minimal loss of reagent from this process, set the pipette volume to 84µl before extracting the reagent from each of the tubes. Vortex briefly.

The PCR reaction mix is 5µl of template/extracted sample elute, or standard and 20µl of the re-hydrated master mix/probe/primer.

Set up samples as outlined below:

1. Pipette 20µl rehydrated mastermix into each respective PCR reaction well or individual tube.
2. Pulse vortex the sample to ensure the material is homogenous just prior to loading. For the standard, pulse vortex then briefly spin down.
3. Add 5µl of respective sample/standard into each respective PCR reaction well or individual tube.
4. Seal the plate or PCR tubes and mix at low speed on the appropriate plate or tube spinner.

After running and setting up the standard curve which is only necessary with each new CyanoDTec kit lot and/or new laboratory technician or operator, its recommended to run at least one of the standards as a positive control and a water (non-template control NTC) in each subsequent run as Quality Control.

Program the PCR machine using the following parameters

The fluorophores/dyes are:

Total Cyanobacteria Test

Total Cyanobacteria (16S rRNA gene)	FAM (495/516nm)
Internal Amplification Control (IAC)	CalFluor Orange (CFO - 538/559nm).

Cyano Toxin Targets

Microcystin/Nodularin (mcyE/ndaF)	FAM (495/516nm).
Cylindrospermospin (cyrA)	CalFluor Orange (CFO - 538/559nm),
Saxitoxin (Sxt A)	CalFluor Red (CFR - 590/610nm)

Cycling Conditions:

The CyanoDTec Assay has been successfully run on many different qPCR platforms. While the parameters listed below work best on most instruments, cycling conditions may require optimization for your instrument. These parameters include Annealing Temperature, Time and Ramp Rate. Please contact Phytoxigene for support documentation and recommendations to specific instruments.

Initial denaturation	95 °C, 2 min
Denaturation	95 °C, 15 s
Annealing- Extension	60°C, 45 s (optics ON)
Ramp Rate	Standard*(up to 2o/sec
Cycle number	40

*Do not run in Fast Mode on Thermo ABI platforms.

Sample Layouts for Phytoxigene™ CyanoDTec Runs

Screening Run and Standard Curve set up to determine the presence of total Cyanobacteria

Cyano 16s/IAC Test Kit

	A	B	C	D	E
1	1,000,000 std, NA026	Sample 3	Sample 11	Sample 19	Sample 27
2	100,000 std, NA015	Sample 4	Sample 12	Sample 20	Sample 28
3	10,000 std, NA014	Sample 5	Sample 13	Sample 21	Sample 29
4	1,000 std, NA013	Sample 6	Sample 14	Sample 22	Sample 30
5	100 std, NA012	Sample 7	Sample 15	Sample 23	Sample 31
6	Zero (0) Blank NTC	Sample 8	Sample 16	Sample 24	Sample 32
7	Sample 1	Sample 9	Sample 17	Sample 25	Sample 33
8	Sample 2	Sample 10	Sample 18	Sample 26	Sample 34

Screening Run and Standard Curve set up to determine the presence of any relevant Toxin Gene

Toxin Gene Test Kit

	A	B	C	D	E
1	1,000,000 std, NA026	Sample 3	Sample 11	Sample 19	Sample 27
2	100,000 std, NA015	Sample 4	Sample 12	Sample 20	Sample 28
3	10,000 std, NA014	Sample 5	Sample 13	Sample 21	Sample 29
4	1,000 std, NA013	Sample 6	Sample 14	Sample 22	Sample 30
5	100 std, NA012	Sample 7	Sample 15	Sample 23	Sample 31
6	Zero (0) Blank NTC	Sample 8	Sample 16	Sample 24	Sample 32
7	Sample 1	Sample 9	Sample 17	Sample 25	Sample 33
8	Sample 2	Sample 10	Sample 18	Sample 26	Sample 34

Simultaneous Screen and standard curve set up for total cyanobacteria and toxin detection on same run

	Cyano 16s/IAC Test Kit		Toxin Gene Test Kit	
	A	B	D	E
1	1,000,000 std, NA026	Sample 3	1,000,000 std, NA026	Sample 3
2	100,000 std, NA015	Sample 4	100,000 std, NA015	Sample 4
3	10,000 std, NA014	Sample 5	10,000 std, NA014	Sample 5
4	1,000 std, NA013	Sample 6	1,000 std, NA013	Sample 6
5	100 std, NA012	Sample 7	100 std, NA012	Sample 7
6	Zero (0) Blank NTC	Sample 8	Zero (0) Blank, NA011	Sample 8
7	Sample 1	Sample 9	Sample 1	Sample 9
8	Sample 2	Sample 10	Sample 2	Sample 10

If there is no change in measuring equipment, operator or reagent lot used, then subsequent runs can use the standard curve generated in previous runs

	Cyano 16s/IAC Test Kit		Toxin Gene Test Kit	
	A	B	E	F
1	16s rRNA +Control or std	Sample 7	toxin +Control or std	Sample 7
2	Blank/NTC	Sample 8	Blank/NTC	Sample 8
3	Sample 1	Sample 9	Sample 1	Sample 9
4	Sample 2	Sample 10	Sample 2	Sample 10
5	Sample 3	Sample 11	Sample 3	Sample 11
6	Sample 4	Sample 12	Sample 4	Sample 12
7	Sample 5	Sample 13	Sample 5	Sample 13
8	Sample 6	Sample 14	Sample 6	Sample 14

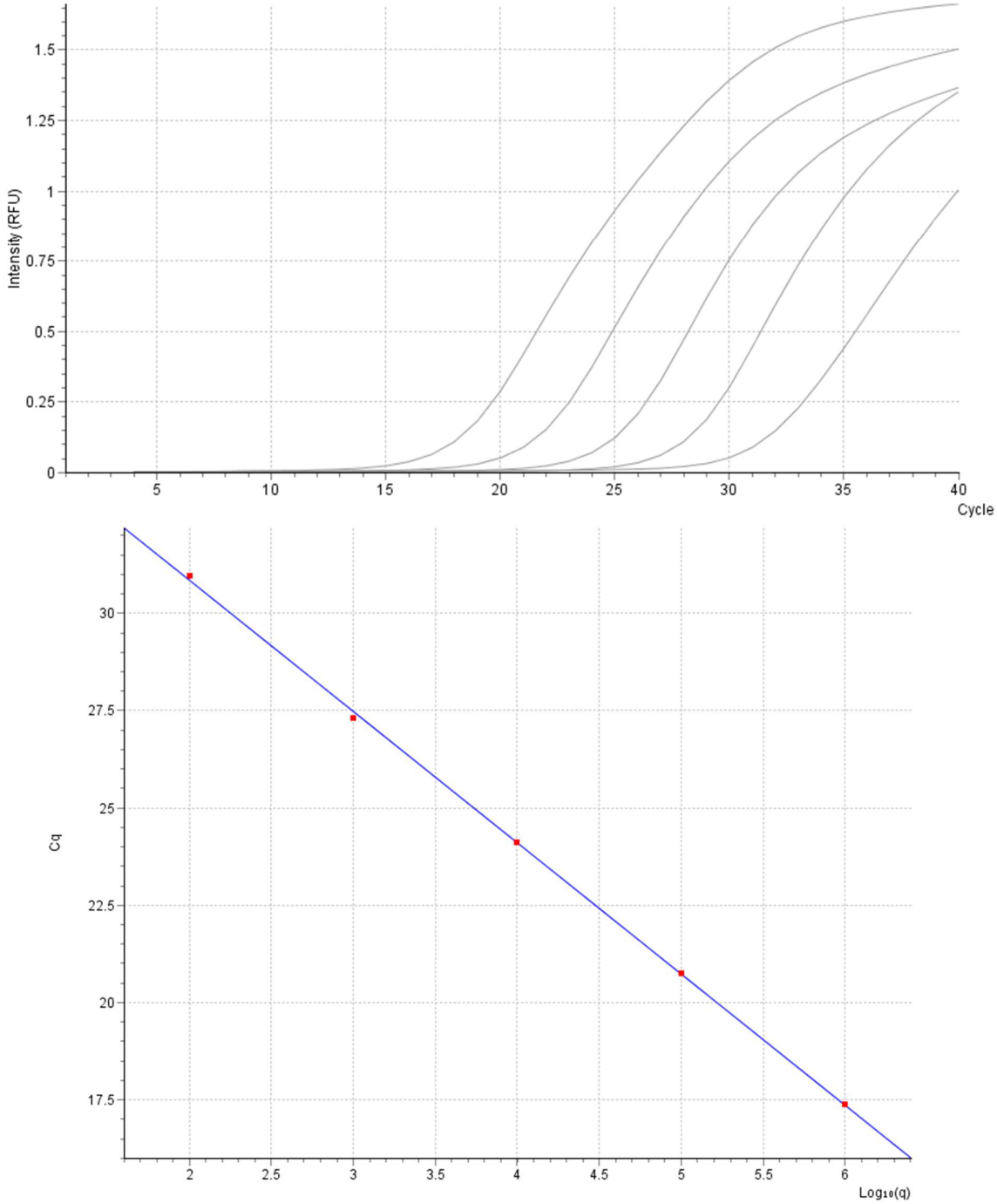
For the 16S rRNA assay, consider any sample not detected (ND) when the sample CT value is more than 4 cycles above the lowest standard. It is not uncommon to get some low level amplification in water at around 37 to 40 cycles, this is well below the limit of detection and it has no impact on the overall assay result.

The IAC Cycle Threshold (CT) will be between 29-32 cycles but varies depending on PCR system used. In any samples where the CT of the IAC is more than 1.5 cycles above the IAC CT of the Blank or Non Template Control, the test result for that relevant sample should be considered invalid and will need to be rerun to exclude the potential of inhibition in the amplification process. Dilute your sample at a minimum of 1:10 and rerun sample.

Indicative threshold CT's and reproducibility for the CyanoNAS standards
(CT values will vary from instrument to instrument)

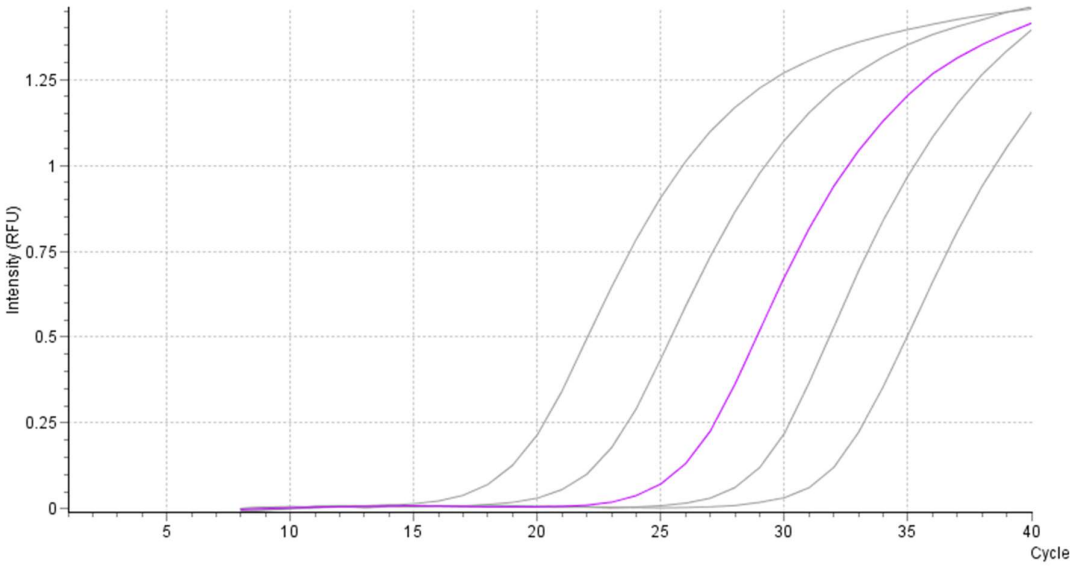
National Measurement Institute reference material B170731	NA026	NA015	NA014	NA013	NA012
Number of molecules per ul ddPCR based	206,000	21,300	2,080	204	20
Nominal number of molecules per qPCR reaction	1,030,000	106,500	10,400	1,020	99
Log nominal number of molecules per qPCR reaction	6	5	4	3	2
Observed CT value qPCR 16S FAM fluorofore	19	22.21	25.72	29.11	32.96
	18.94	22.45	25.66	29.39	32.8
	19.04	22.1	25.81	29.17	32.74
Average	**18.99**	**22.25**	**25.73**	**29.22**	**32.83**
SD	0.05	0.18	0.08	0.15	0.11
Observed CT value qPCR McyE FAM fluorofore	18.94	22.05	25.37	28.97	32.48
	18.51	22.11	25.53	29.05	32.13
	18.56	22.29	25.13	28.45	32.34
Average	**18.67**	**22.15**	**25.34**	**28.83**	**32.32**
SD	**0.24**	**0.12**	**0.2**	**0.33**	**0.18**
Observed CT value qPCR CryA CY3/HEX/Cal 560 fluorofore	19.17	22.42	25.73	29.12	32.03
	19.07	22.5	25.88	28.94	32.10
	18.93	22.27	25.5	28.96	31.94
Average	**19.06**	**22.4**	**25.7**	**29.01**	**31.99**
SD	0.12	0.12	0.19	0.10	0.05
Observed CT value qPCR SxtA ROX/TexasRed/Cal 610 fluorofore	18.96	22.24	25.64	28.99	32.35
	18.89	22.27	25.73	28.87	32.24
	18.77	22.21	25.38	28.85	32.37
Average	**18.87**	**22.24**	**25.58**	**28.9**	**32.32**
SD	0.1	0.03	0.18	0.08	0.07

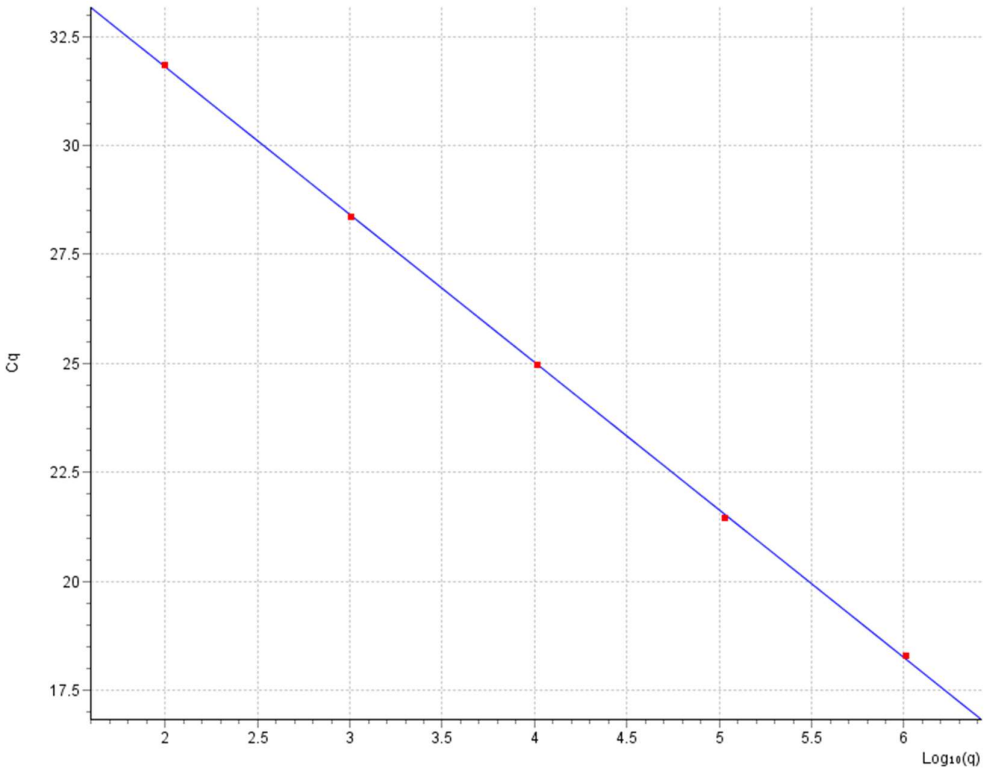
Indicative cycling curves for each of the targets and standards.



Cq=-3.37 log10(q)=+37.57 E=0.98% r^2=0.9997

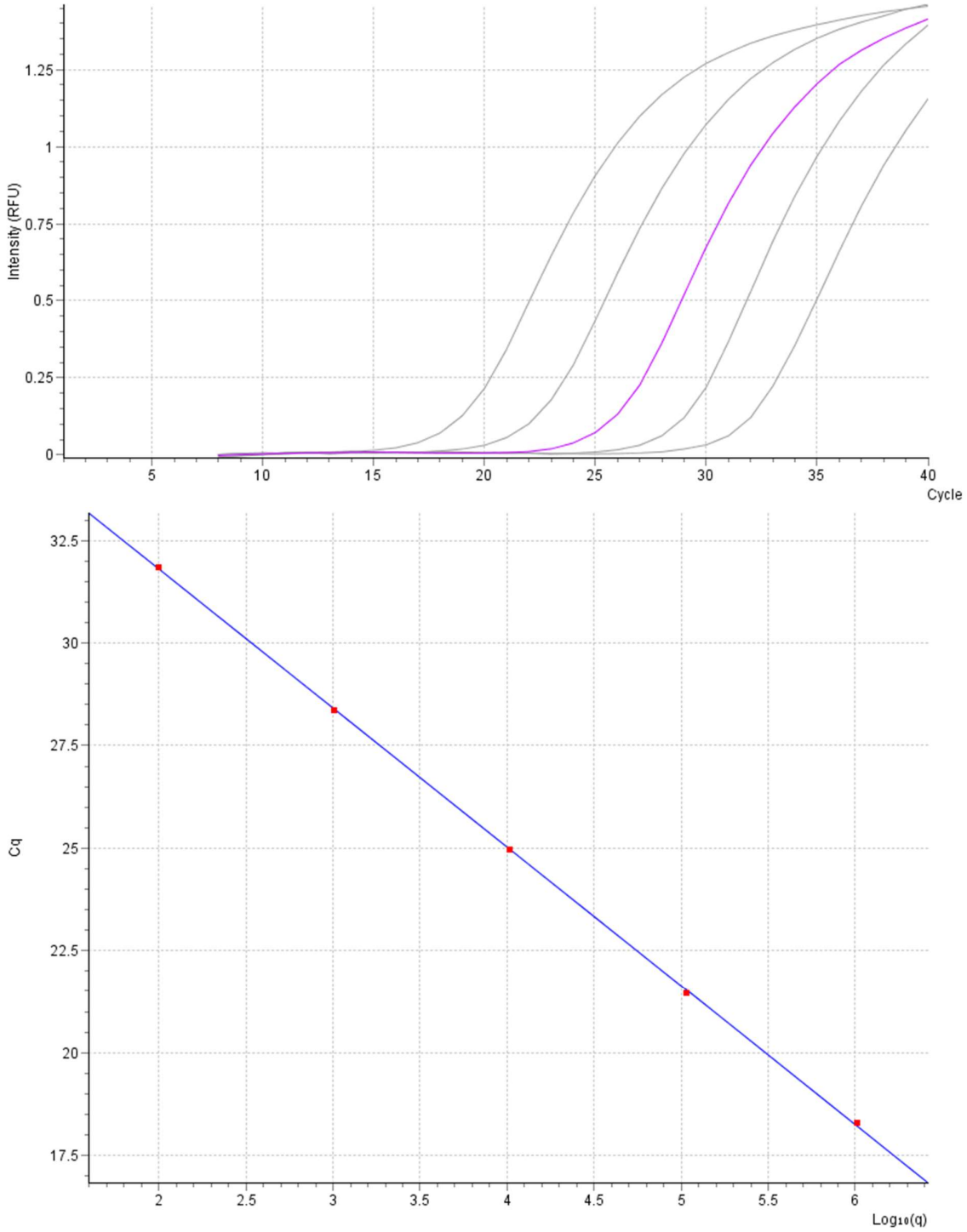
Figure 1. Amplification plot and standard curve for total cyanobacteria (16SrRNA) assays using CyanoDTec kit on a MyGoPro





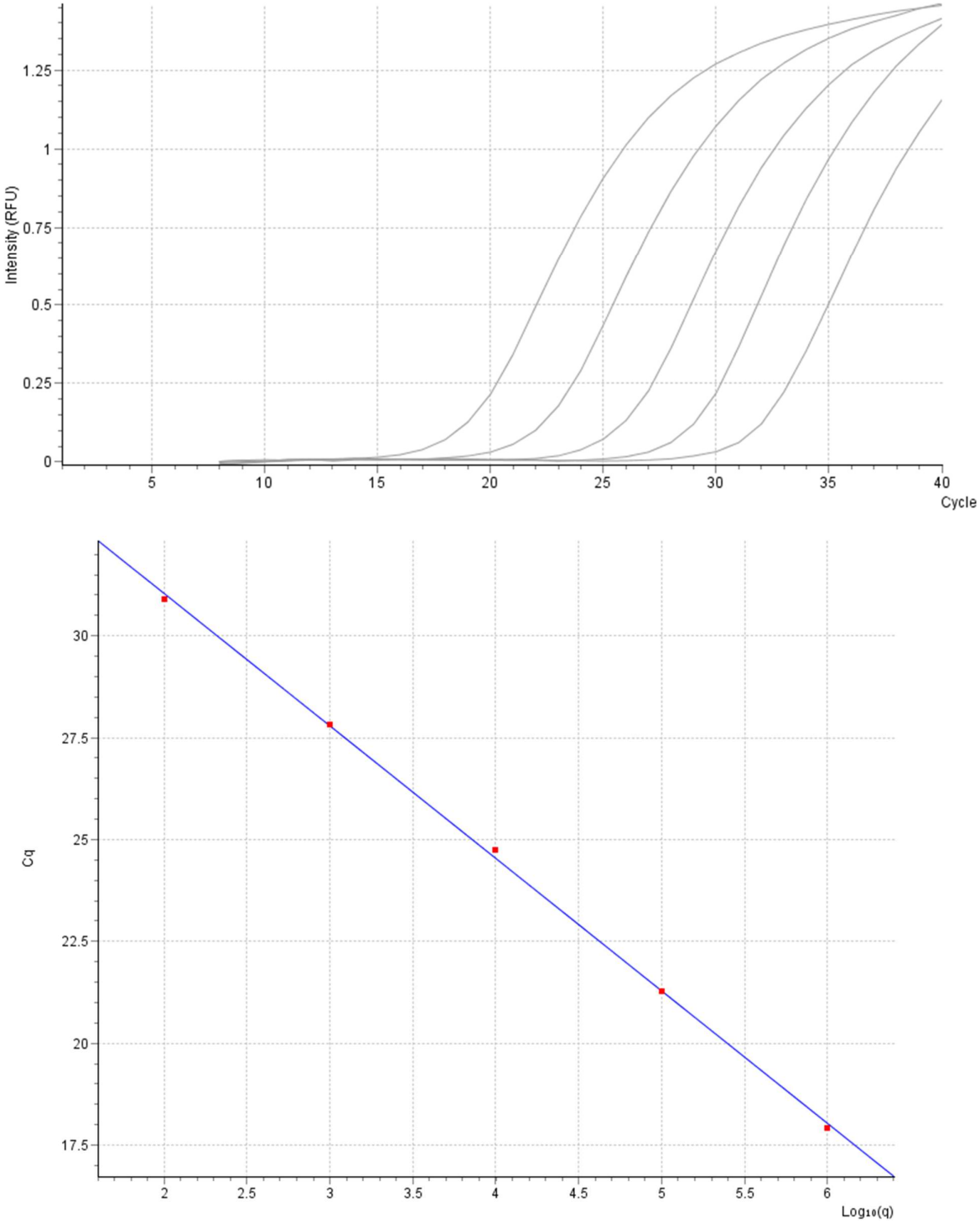
Cq=-3.26 log10(q)=+37.69 E=1.02% r^2=0.9989

Figure 2. Amplification plot and standard curve for microcystin and nodularin (mcyE/ndaF) target assays using CyanoDTec kit on a MyGoPro



Cq=-3.39 log10(q)=+38.58 E=0.97% r^2=0.9998

Figure 3. Amplification plot and standard curve for cylindrospermospin (cyrA) target assays using CyanoDTec kit on a MyGoPro



Cq=-3.25 log10(q)=+37.52 E=1.03% r^2=0.9998

Figure 4. Amplification plot and standard curve for saxitoxin (sxtA) target assays using CyanoDTec kit on a MyGoPro

SENSITIVITY AND SPECIFICITY

The assay is capable with 95% degrees of confidence, of detecting accurately down to 45 copies per reaction. The resulting accuracy for your test is dependent on starting volume and extraction protocol and yield.

The assay and targets have been tested against a wide range of species and strains both from culture stocks and environmental samples. It has been shown to be highly sensitive and specific as it relates to discriminating toxin and non-toxin producing cyanobacteria.

Species	Strain	Hepatoxic	Hepatoxin	Other Cyanotoxins	Origin
Chroococcales					
Microcystis aeruginosa	PCC7806	+	microsystin	-	The Netherlands
Microcystis aeruginosa	PCC7005	+	microsystin	-	Scotland
Microcystis sp.	UTEX 2667	+	microsystin	-	United States
Microcystis sp.	UTEX 2664	+	microsystin	-	United States
Microcystis viridis	NIES 102	+	microsystin	-	Japan
Microcystis wesenbergii	NIES 107	+	microsystin	-	Japan
Microcystis aeruginosa	HUB53	-	-	-	Germany
Microcystis aeruginosa	UWOCC C1	-	-	-	United States
Microcystis aeruginosa	UWOCC C4	-	-		United States
Microcystis aeruginosa	UWOCC P3		-		United States
Microcystis sp.	UWOCC Bauld B	-	-	-	Australia
Synechocystis sp.	PCC6803	-	-	-	United States
Nostocales					
Anabaena sp.	strain 202A2	+	microcystin	-	Finland
Anabaena circinalis	NIES 19	-	-	saxitoxin	England
Cylindrospermopsis sp.	AWT	-	-	cylindrospermopsin	Australia
Nodularia harveyana	PCC7804	+	nodularin	-	France
Nodularia harveyana	PCC73104	-	-	-	Canada
Notularia spumigena	NROS 10	+	nodularia	-	Finland
Notularia spumigena	BY1	+	nodularia	-	Baltic Sea
Notularia spumigena	HEM	+	nodularia	-	Baltic Sea
Nostoc sp.	strain 152	+	microcystin	-	Japan
Nostoc sp.	PCC7120	-	-	-	United States
Nostoc sp.	PCC73120	-	-	-	Australia
Oscillatoriales					
Lyngbya sp.	BAN 01	-	-	n/a	Australia
Oscillatoria sp.	strain18R	+	microcystin	-	Finland
Oscillatoria sp.	*strain* 195	+	microcystin	-	Finland
Phormidium tenue	-	-	-	n/a	Japan
Phormidium sp.	isolate 2-26b	+	microcystin	n/a	United States
Phormidium sp.	isolate 1-6c	+	microcystin	n/a	United States
Phormidium sp.	isolate 4-19b	+	microcystin	n/a	United States

Table 1. Toxin and Non-Toxin cyanobacteria species distribution

Cyanobacteria Strains	Toxicity	SxtA	CYLN	MCY	16S rRNA
A.circinalis AWQC118C	SXT	+	-	-	+
A.circinalis AWQC131C	SXT	+	-	-	+
A.circinalis AWQC134C	SXT	+	-	-	+
A.circinalis AWQC150E	SXT	+	-	-	+
A.circinalis AWQC271C	-	-	-	-	+
A.circinalis AWQ306A	-	-	-	-	+
A.circinalis AWQC310F	-	-	-	-	+
A.circinalis AWQC342D	-	-	-	-	+
Aph.flos-aquaea NH-5	SXT	+	-	-	+
Aph.ovalisporum APH028A	CLYN	-	+	-	+
C.raciborskii T3	SXT	+	-	-	+
C.raciborskii 23B	CLYN	-	+	-	+
C.raciborskii GOON	CLYN	-	+	-	+
C.raciborskii GERM1	-	-	-	-	+
C.raciborskii MARAU1	-	-	-	-	+
L.wollei	SXT	+	-	-	+
N.Spumigena NSOR10	NOD	-	-	+	+
M.aeruginosa PCC7806	MCY	-	-	+	+
C.raciborskii AWT205	CLYN	-	+	-	+
Nostoc sp. PCC7120	-	-	-	-	+
N.punctiforme PCC73102	-	-	-	-	+
SXT, Saxitoxin; CLYN, cylindrospermopsin; MCY, Microcystin; NOD, Nodularin; +, gene amplified					

Table 2. Cyanobacteria strains and species tested using target genes

			Specificity of primers for;				
#	Cyanobacterial Strain	Produces microcystin	Anabaena mcyE	Microcystis mcyE	Planktothrix mcyE	Cyanobacteria 16S rRNA	References
1	*Anabaena sp.* 66A	+	+	-	-	NT	Vaitomaa et al.,2003
2	*Anabaena sp.* 90	+	+	-	-	NT	Vaitomaa et al.,2003
3	*Anabaena sp*. 202A1	+	+	-	-	NT	Vaitomaa et al.,2003
4	*Anabaena sp.* 318	+	+	-	-	NT	Vaitomaa et al.,2003
5	*A. Lemmermannii* NIVA CYA 83/1	+	+	-	-	+	Vaitomaa et al.,2003
6	*Anabaena sp.* 86	-	-	-	-	+	Vaitomaa et al.,2003
7	*Anabaena sp.* 14	-	-	-	-	+	Vaitomaa et al.,2003
8	*Anabaena sp*. CPCC 426	-	-	-	-	+	Ngwa et al., 2012
9	*A. circinalis* LMECYA 175	-	-	-	-	+	Valerio et al.,2010
10	*A. planctonica* LMECYA 177	-	-	-	-	+	Valerio et al.,2010
11	*A. spiroides* LMECYA 161	-	-	-	-	+	Valerio et al.,2010
12	*A. flos aquae* CPCC 67	-	-	-	-	+	Ngwa et al., 2012
13	*Aphanocapsa rivularis* CPCC 596	-	-	-	-	+	Ngwa et al., 2012
14	*Aphanothece flocculosa* CPCC 597	-	-	-	-	+	Ngwa et al., 2012
15	*Aph. Gracile* LMECYA 64	NT	-	-	-	+	
16	*Aph. Flos aquae* LMECYA 88	NT	-	-	-	+	
17	*Aph. Issatschenkoi* LMECYA 166	NT	-	-	-	+	
18	*C. raciborskii* LMECYA 135	NT	-	-	-	+	
19	*Planktothrix sp.* 49	+	-	-	+	-	Vaitomaa et al.,2003
20	*Planktothrix sp.* 97	+	-	-	+	+	Vaitomaa et al.,2003
21	*Planktothrix sp.* 213	+	-	-	+	+	Vaitomaa et al.,2003
22	*P. prolifica* NIVA CYA 128	+	-	-	+	+	Vaitomaa et al.,2003
23	*P. agardhii* NIVA CYA 127	+	-	-	+	+	Vaitomaa et al.,2003
24	*Planktothrix sp.* 45	-	-	-	-	NT	Vaitomaa et al.,2003
25	*M. aeruginosa* LMECYA 59	+	-	+	-	+	Valerio et al.,2010
26	*M. aeruginosa* LMECYA 87	+	-	+	-	+	Valerio et al.,2010
27	*M. aeruginosa* CPCC 299	+	-	+	-	+	Ngwa et al., 2012
28	*M. aeruginosa* CPCC 300	+	-	+	-	+	Ngwa et al., 2012
29	*M. aeruginosa* LMECYA 144	-	-	-	-	+	Valerio et al.,2010
30	*M. aeruginosa* LMECYA 183	-	-	-	-	+	Valerio et al.,2010
31	*M. aeruginosa* CPCC124	-	-	-	-	+	Ngwa et al., 2012
32	*M. aeruginosa* CPCC468	-	-	-	-	+	Ngwa et al., 2012
33	*M. aeruginosa* CPCC632	-	-	-	-	+	Ngwa et al., 2012
34	*M. aeruginosa* CPCC633	-	-	-	-	+	Ngwa et al., 2012
35	*M. flos aquae* CPCC461	-	-	-	-	+	Ngwa et al., 2012
36	*M. aeruginosa* CPCC459	NT	-	-	-	+	
37	*S. leopoliensis* CPCC102	-	-	-	-	+	Ngwa et al., 2012

a Studies from which microcystin production (+) or lack of production (-) were inferred.

b Production (+) or non-production (-) of microcystin synthetase (McyE) or cyanobacterial 166 rRNA amplicons by the respective mycE and NT; not tested

Table 3. Cyanobacteria strains used for McyE primer sensitivity and specificity testing

INTERPRETATION OF RESULTS

- The presence of a respective target toxin gene does not immediately infer that the toxin will be present, but is highly suggestive that there is, and monitoring frequency should be increased. Additional testing options should also be considered.
- The level of genes present is not directly correlative to potential toxin present but recent data would suggest this to be the case.
- A negative result does not immediately infer that there are no toxins present, Apart from the potential presence of extracellular toxins that naturally diffuse through a water system, it's also known that when cyanobacteria blooms collapse there is a release of their intracellular toxin, test results in these cases will be negative for the respective toxin genes.
- At this stage specific gene levels, that may reflect the need for raising alert warnings, have not been set. User assessment based on other parameters and experience should be considered.
- While every effort has been made to test all known toxin producers to insure both the specificity and sensitivity of the assay, there is no guarantee that a species that produces one of the target toxins is not detected by the assay. There is likely to be an array of potential toxin producing species not yet characterised.

Limitations in the interpretation of results need to be considered into any recommendation or report issues by the laboratory.

LIMITATIONS

The assay has been designed to meet precise analytical standards required for detection of molecular targets responsible for the production of specific cyanobacteria toxins.
Sampling techniques and locations can result in highly variable results. Standard operating Processes should be developed and followed when assessing the requirements for collecting and processing of environmental samples
The assay is highly contingent on the efficiency of the extraction process used. This procedure outlines three methods that have shown work and are fit for purpose. Any result should be conditional on the parameters known at the time, which include the extraction and sampling procedures used.

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Australian Patent no. 2006254717, Detection of hepatotoxic cyanobacteria
European Patent no. 1888766 Detection of hepatotoxic cyanobacteria.
United States patent no.US8927217 Detection of hepatotoxic cyanobacteria
South African patent no. 2007/10929 Detection of hepatotoxic cyanobacteria
Japanese Patent No. 5722765 Cyanobacteria saxitoxin gene cluster and detection of cyanotoxic organisms
South African patent no. 2010/07749 Cyanobacteria saxitoxin gene cluster and detection of cyanotoxic organisms
PCT/AU2012/000541; Cyanobacteria saxitoxin gene cluster and detection of cyanotoxic organisms
European Patent no.2279265 Cyanobacteria saxitoxin gene cluster and detection of cyanotoxic organisms
United States Patent Application No. 12/989394 Cyanobacteria saxitoxin gene cluster and detection of cyanotoxic organisms

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Summary of the CyanoDTec Procedure

Sample Preparation

1. Mix sample thoroughly, may require at least 30 seconds of shaking to break up clumps of cyanobacteria. Sample may need to be diluted if dense with cells.
2. Draw 10ml into syringe and filter through filter holder containing a 1 or 3 um filter
3. Place filter into the bead lysis tube, face up
4. Place on horizontal tube holder and vortex at maximum speed for a minimum of 15 minutes
5. While samples are being vortexed, rehydrate appropriate amount of master mix
6. Pipette 20ul of master mix into respective (Total Cyanobacteria 16s and Toxin gene) reaction tubes
7. Upon completion of the sample vortexing, spin the samples in the micro-centrifuge for 30 seconds
8. Immediately pipette 5µl of sample into respective 16s and/or toxin gene reaction tubes
9. Spin reaction tubes
10. Place in thermocycler and initiate run